Time Series Modeling & Decision-Making Report

Data file: synthetic_monthly_sales_2019_2023.csv

Date range: 2019-01-01 to 2023-12-01

Objective: Forecast next periods and provide prescriptive recommendations (safety stock & production).

Observations: 60

Model used: Holt-Winters (additive trend & additive seasonality, seasonal_periods=12)

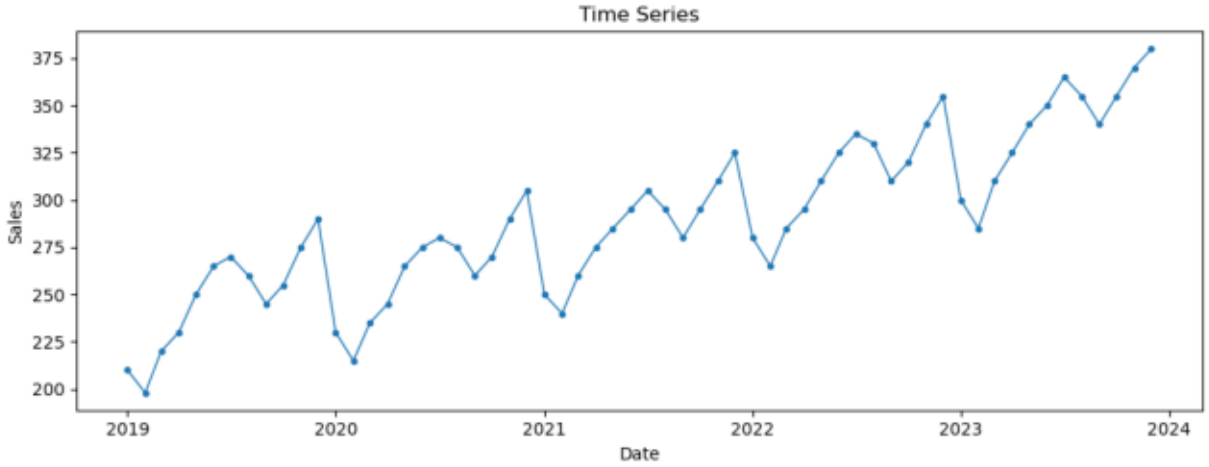
Holdout (last 12) RMSE: 5.75

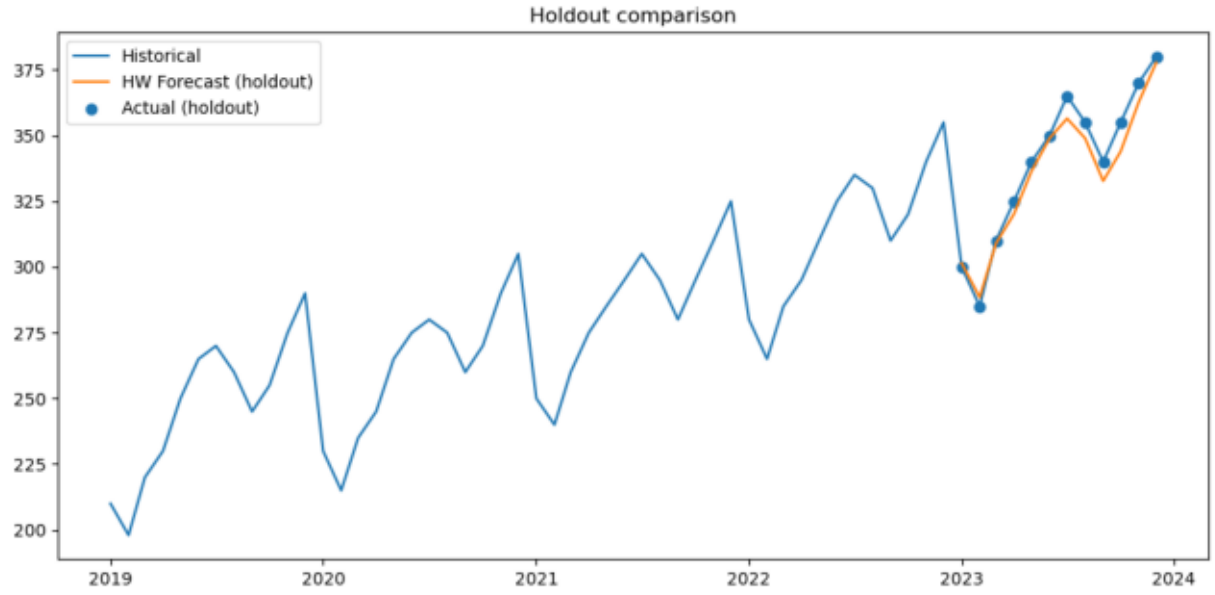
Holdout (last 12) MAPE: 1.40%

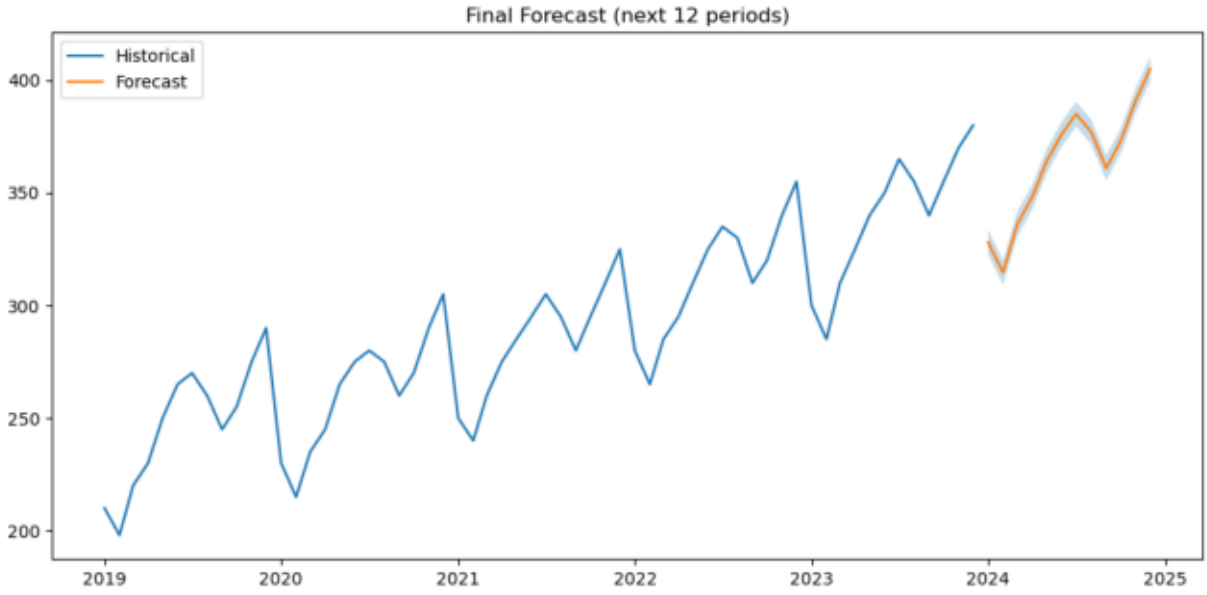
Residual std (in-sample): 2.80

Safety stock (z~1.645 * resid_std): 4.61 (units)

Costs used (placeholders): holding_cost_per_unit_per_period=0.5, stockout_cost_per_unit=5.0


Time Series
Sales
375
350
325
300
275
250
225
200
2019
2020
2021
2022
2023
2024
Date


Holdout comparison
Historical
HW Forecast (holdout)
Actual (holdout)
375
350
325
300
275
250
225
200
2019
2020
2021
2022
2023
2024


Final Forecast (next 12 periods)
Historical
Forecast
400
350
300
250
200
2019
2020
2021
2022
2023
2024
2025

Decision Table (next periods)

2024-03-01	336.03	330.54	341.53	341.0	0.04	0.04
2024-04-01	348.03	342.54	353.53	353.0	0.04	0.04
2024-05-01	364.03	358.54	369.53	369.0	0.04	0.04
2024-06-01	376.03	370.54	381.53	381.0	0.04	0.04
2024-07-01	385.03	379.54	390.53	390.0	0.04	0.04
2024-08-01	377.03	371.54	382.53	382.0	0.04	0.04
2024-09-01	361.03	355.54	366.53	366.0	0.04	0.04
2024-10-01	373.03	367.54	378.53	378.0	0.04	0.04
2024-11-01	391.03	385.54	396.53	396.0	0.04	0.04
2024-12-01	405.03	399.54	410.53	410.0	0.04	0.04